

EXP 22:

Perform Sharpening of Image using Laplacian mask with positive center coefficient.

0	-1	0
-1	5	-1
0	1	0

Mask of Laplacian + addition

$$\begin{aligned}g(x, y) &= f(x, y) - [f(x+1, y) + f(x-1, y) \\&\quad + f(x, y+1) + f(x, y-1) + 4f(x, y)] \\&= 5f(x, y) - [f(x+1, y) + f(x-1, y) \\&\quad + f(x, y+1) + f(x, y-1)]\end{aligned}$$

AIM:

Sharpening of Image using Laplacian mask with positive center coefficient.

Program:

```
import cv2

import numpy as np

img = cv2.imread(r'C:\Users\pavan\OneDrive\Opencv\x.jpg')

if img is None:

    print("Error: Image not found or path is incorrect")

else:

    kernel = np.array([[[-1, -1, -1],
                        [-1, 8, -1],
                        [-1, -1, -1]]])

    sharpened = cv2.filter2D(img, -1, kernel)

    cv2.imshow('Original Image', img)

    cv2.imshow('Sharpened Image (Positive Laplacian)', sharpened)

    cv2.imwrite('Sharpened_Image_Exp22.jpg', sharpened)

    cv2.waitKey(0)

    cv2.destroyAllWindows()
```

INPUT:



Output:



RESULT:- Thus, the program is executed successfully.